IMPLEMENTATION OF PURE-FLUID CUBIC EQUATIONS OF STATE AND RESIDUAL PROPERTY SOLVERS

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***Abstract*—This project report presents the rigorous engineering formulation, mathematical architecture, numerical root-solving, and software implementation of a unified desktop thermodynamics application. The computational engine calculates compressibility factors (Z), molar volumes (V), fugacity coefficients (φ), and residual departure functions (H^R, S^R) for ten industrially vital pure substances across wide pressure-temperature regimes. Four prominent cubic equations of state—van der Waals (vdW), Redlich-Kwong (RK), Soave-Redlich-Kwong (SRK), and Peng-Robinson (PR)—are modeled. A closed-form trigonometric Cardano solver determines the real polynomial roots, while equifugacity flash iterations capture vapor-liquid equilibria (VLE) without convergence failure. To guarantee standalone portability without external package management, the user interface and custom coordinate-mapped data visualization engines are engineered entirely using standard Python and Tkinter primitives. Quantitative comparisons against empirical benchmarks demonstrate the superior accuracy of the two-parameter Peng-Robinson EOS, specifically for subcritical polar and asymmetric hydrocarbon media.*

I. INTRODUCTION & GAS SET SELECTION

Accurate thermodynamic phase and volumetric calculations are fundamental to chemical plant synthesis, pipeline fluid mechanics, separator sizing, and compressor design. In industrial practice, operating conditions deviate substantially from the ideal gas assumption (PV = RT) due to molecular co-volume displacement and intermolecular attraction. Cubic equations of state (CEOS) offer an optimal balance between algebraic tractability and physical reliability.

To address diverse industrial applications, the software curates ten representative fluids categorized into structural classes:

- **Light Hydrocarbons (Alkanes & Alkenes):** Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8), n-Butane (n-C_4H_{10}), and Ethylene (C_2H_4). These characterize natural gas networks, cryogenics, and olefin polymerization units.
- **Atmospheric & Inorganic Gases:** Nitrogen (N_2), Oxygen (O_2), and Carbon Dioxide (CO_2). These represent air separation facilities, supercritical extraction circuits, and carbon capture networks.
- **Supercritical / Non-Polar Media:** Hydrogen (H_2), an energy carrier exhibiting quantum mechanical dispersion characteristics and low inversion temperatures.
- **Strongly Associating / Polar Media:** Water (H_2O), featuring extensive directional hydrogen-bonding networks and anomalous critical properties.

The thermophysical baseline constants for all ten compounds—comprising critical temperature (T_c), critical pressure (P_c), Pitzer acentric factor (ω), and the four-term empirical ideal-gas isobaric heat capacity parameters (A, B, C, D)—are summarized in Table I.

Compound	T_c (K)	P_c (bar)	ω	A	B×10³	C×10⁶	D×10⁻⁵
Methane	190.6	45.99	0.012	1.702	9.081	−2.164	0.000
Ethane	305.3	48.72	0.100	1.131	19.225	−5.561	0.000
Propane	369.8	42.48	0.152	1.213	28.785	−8.824	0.000
n-Butane	425.1	37.96	0.200	1.935	36.915	−11.402	0.000
Ethylene	282.3	50.40	0.087	1.424	14.394	−4.392	0.000
Nitrogen	126.2	34.00	0.038	3.280	0.593	0.000	0.040
Oxygen	154.6	50.43	0.022	3.639	0.506	0.000	−0.227
Carbon Dioxide	304.2	73.83	0.224	5.457	1.045	0.000	−1.157
Hydrogen	33.19	13.13	−0.216	3.249	0.422	0.000	0.083
Water	647.1	220.55	0.345	3.470	1.450	0.000	0.121

II. UNIFIED THERMODYNAMIC FRAMEWORK

Rather than implementing four disconnected equations, the codebase utilizes the generalized two-parameter cubic equation of state framework. Expressed in pressure-explicit form:

$$P = RT / (V - b) - a(T) / [(V + \varepsilon b)(V + \sigma b)]$$

Here, b models the finite repulsive co-volume of the molecules, while a(T) quantifies attractive intermolecular forces modulated by temperature. At the critical point, the inflection criteria $(\partial P/\partial V)_{Tc} = 0$ and $(\partial^2 P/\partial V^2)_{Tc} = 0$ yield the universal scale relations:

$$b = \Omega_b R T_c / P_c$$
$$a(T) = \Psi_a [R^2 T_c^2 / P_c] \alpha(T_r, \omega)$$

Dimensionless reduced variables transform the equation into a cubic polynomial in terms of the compressibility factor Z = PV/(RT):

$$\beta = \Omega_b P_r / T_r = b P / (RT)$$
$$q = \Psi_a \alpha(T_r, \omega) / [\Omega_b T_r] = a(T) / (b R T)$$
$$Z^3 + c_2 Z^2 + c_1 Z + c_0 = 0$$

The polynomial coefficients c_2, c_1, and c_0 are strictly functions of the model geometry constants (σ, ε) and dimensionless groups (β, q):

$$c_2 = \beta(\sigma + \varepsilon) - (1 + \beta)$$
$$c_1 = \sigma\varepsilon\beta^2 - (1 + \beta)(\sigma + \varepsilon)\beta + q\beta$$
$$c_0 = -(1 + \beta)\sigma\varepsilon\beta^2 - q\beta^2$$

The structural parameters corresponding to each implemented model are detailed in Table II:

Model	σ	ε	Ω_b	Ψ_a	$\alpha(T_r, \omega)$ Form
vdW (1873)	0	0	0.12500	0.42188	1.0
RK (1949)	1	0	0.08664	0.42748	$T_r^{-0.5}$
SRK (1972)	1	0	0.08664	0.42748	$[1 + m(1 - T_r^{0.5})]^2$
PR (1976)	$1+\sqrt{2}$	$1-\sqrt{2}$	0.07780	0.45724	$[1 + \kappa(1 - T_r^{0.5})]^2$

For the Soave-Redlich-Kwong and Peng-Robinson equations, the empirical correction factors m and κ adapt the attractive parameter to non-spherical molecular geometry via Pitzer's acentric factor ω:

$$m = 0.480 + 1.574\ \omega - 0.176\ \omega^2$$
$$\kappa = 0.37464 + 1.54226\ \omega - 0.26992\ \omega^2$$

III. CLOSED-FORM ANALYTICAL ROOT SOLVING

Numerical root finders (such as Newton-Raphson) frequently diverge or cycle near phase boundaries. To ensure unconditional robustness, this application solves the depressed cubic equation analytically using Cardano's formulation. Substituting $Z = x - c_2/3$ transforms the cubic into depressed form $x^3 + px + r = 0$, where:

$$p = c_1 - c_2^2 / 3$$
$$r = c_0 - c_1\ c_2 / 3 + 2\ c_2^3 / 27$$

Evaluating the polynomial discriminant via intermediate parameters $Q = -p/3$ and $R = r/2$:

$$Q = (c_2^2 - 3\ c_1) / 9$$
$$R = (2\ c_2^3 - 9\ c_2\ c_1 + 27\ c_0) / 54$$
$$D = R^2 - Q^3$$

Two distinct mathematical regimes govern the physical roots:

- **Supercritical / Monophasic Domain (D ≥ 0):** The equation possesses exactly one real root, evaluated via hyperbolic cardanic radicals:

$$A = -\ \mathrm{sgn}(R)\ [|R| + \sqrt{(R^2 - Q^3)}]^{1/3}$$
$$B = Q / A \quad (\text{if } A \neq 0, \text{ else } 0)$$
$$Z_1 = (A + B) - c_2 / 3$$

- **Subcritical Two-Phase Domain (D < 0):** The discriminant yields three distinct real roots, evaluated via Viete's trigonometric formulation:

$$\theta = \arccos[\ R / \sqrt{(Q^3)}\ ]$$
$$Z_k = -\ 2\ \sqrt{Q}\ \cos[(\theta + 2\pi k) / 3] - c_2 / 3 \quad (k = 0, 1, 2)$$

Physical acceptability requires $Z > \beta$, as volume must exceed co-volume ($V > b$). Roots violating $Z > \beta$ are discarded. When three valid roots exist, the largest root corresponds to the vapor phase ($Z_{vap} = \max\{Z_k\}$), the smallest to the liquid phase ($Z_{liq} = \min\{Z_k\}$), and the intermediate root represents an unphysical, mechanically unstable state ($(\partial P/\partial V)_T > 0$).

IV. RESIDUAL PROPERTIES & FUGACITY

Departure functions quantify the exact difference between real-fluid properties and ideal-gas behavior at identical temperature and pressure. From fundamental thermodynamic differential relations, the residual Gibbs free energy G^R relates directly to the fugacity coefficient φ:

$$G^R / (RT) = \ln \varphi = \int_0^P [Z - 1]\ (dP / P)$$

Integrating the generalized cubic EOS yields the dimensionless integral I:

$$I = \int_0^1 dz / [(z + \sigma\beta)(z + \varepsilon\beta)]$$
$$I = [1 / (\sigma - \varepsilon)]\ \ln[(Z + \sigma\beta) / (Z + \varepsilon\beta)] \quad (\sigma \neq \varepsilon)$$
$$I = \beta / (Z + \varepsilon\beta) \quad (\text{vdW limit, } \sigma = \varepsilon = 0)$$

The fugacity coefficient φ and fugacity f are subsequently calculated via:

$$\ln \varphi = Z - 1 - \ln(Z - \beta) - q\ I$$
$$f = \varphi\ P$$

Differentiating the temperature-dependent attractive parameter establishes the residual enthalpy H^R and residual entropy S^R:

$$d\ln\alpha / d\ln T_r = (T_r / \alpha)\ (d\alpha / dT_r)$$
$$H^R / (RT) = Z - 1 + [(d\ln\alpha / d\ln T_r) - 1]\ q\ I$$
$$S^R / R = \ln(Z - \beta) + (d\ln\alpha / d\ln T_r)\ q\ I$$

For SRK and PR models, the logarithmic derivative is derived explicitly:

$$d\ln\alpha / d\ln T_r = -\ \kappa\ \sqrt{T_r}\ /\ [1 + \kappa\ (1 - \sqrt{T_r})]$$

V. VAPOR-LIQUID EQUILIBRIUM & SATURATION

Phase stability between coexisting liquid and vapor phases requires equality of temperature, pressure, and chemical potentials ($\mu_{liq} = \mu_{vap}$), which simplifies to the equifugacity criterion:

$$f_{liq}(T, P^{sat}) = f_{vap}(T, P^{sat}) \quad \Leftrightarrow \quad \varphi_{liq}(T, P^{sat}) = \varphi_{vap}(T, P^{sat})$$

The saturation pressure P^{sat} at an operating subcritical temperature $T < T_c$ is solved via an iterative accelerated Picard substitution loop. The initial estimate is generated from Wilson's empirical correlation:

$$P_{(0)}^{sat} = P_c\ \exp[\ 5.373\ (1 + \omega)\ (1 - T_c / T)\ ]$$

At each iteration step n, the cubic equation is solved for Z_{liq} and Z_{vap}. Fugacity coefficients are evaluated, and the pressure is corrected by the fugacity ratio:

$$P_{(n+1)}^{sat} = P_{(n)}^{sat} \cdot [\ \varphi_{liq} / \varphi_{vap}\ ]$$

Convergence is achieved when $|\varphi_{liq} / \varphi_{vap} - 1.0| < 10^{-9}$. Once saturation is localized, the latent heat of vaporization ΔH_{vap} is determined from the departure differences:

$$\Delta H_{vap} = H_{vap} - H_{liq} = H_{vap}^R - H_{liq}^R$$

VI. ABSOLUTE ENTHALPY & ENTROPY INTEGRATIONS

To compute state-to-state enthalpy (ΔH) and entropy (ΔS) transitions, real-gas properties are anchored to an ideal-gas reference state at T_{ref} = 298.15 K and P_{ref} = 1.0 bar, where H^{ig} = 0 and S^{ig} = 0:

$$H(T, P) = H^{ig}(T) + H^{R}(T, P)$$
$$S(T, P) = S^{ig}(T, P) + S^{R}(T, P)$$

The ideal gas enthalpy and entropy integrals are calculated analytically from the empirical heat capacity polynomial $C_p^{ig} / R = A + B\,T + C\,T^2 + D / T^2$:

$$H^{ig}(T) = R\,[\,A(T - T_0) + (B/2)(T^2 - T_0^2) + (C/3)(T^3 - T_0^3) - D(1/T - 1/T_0)\,]$$
$$S^{ig}(T, P) = R\,[\,A \ln(T/T_0) + B(T - T_0) + (C/2)(T^2 - T_0^2) - (D/2)(1/T^2 - 1/T_0^2)\,] - R \ln(P/P_0)$$

VII. SOFTWARE ARCHITECTURE & UI DESIGN

The application is engineered as a standalone desktop package without requiring external third-party scientific distributions. Key structural choices include:

- **Standard Library Foundation:** Executed strictly via Python's built-in `tkinter`, `ttk`, and `math` modules, preventing deployment hurdles or binary dependency mismatches.
- **Custom Vector Canvas Engine:** Plotting routines map physical coordinates (P, Z, T, H^R) to pixel space on a Tkinter canvas. Tick generators, dynamic axes scalers, grid lines, and Bézier curves render real-time isotherms, EOS comparative bar charts, and phase envelopes natively.
- **Asynchronous UI Resilience:** The window configuration employs top-level window elevation (`root.lift()`) and active focus grabs to ensure reliable presentation across varied desktop window managers.
- **Tabbed Functional Layout:** A split-pane architecture segments inputs into five functional tabs: State Evaluation, State Changes (ΔH, ΔS), Isotherm P-Scans with CSV export, Multi-EOS Comparisons, and VLE Saturation Solvers.

VIII. VERIFICATION & DISCUSSION

Model verification was performed across subcritical, near-critical, and dense supercritical fluid regimes. Table III provides a comparative evaluation of all four equations of state for Carbon Dioxide (CO_2) at T = 320 K (supercritical, T_r = 1.052) and P = 80 bar (P_r = 1.084), benchmarked against empirical NIST Chemistry WebBook data.

EOS Model	Z	V (cm^3/mol)	H^R (J/mol)	φ	Error (%)
vdW	0.5482	182.3	−2,891.4	0.7412	+44.6%
RK	0.4120	137.0	−4,120.8	0.6280	+8.7%
SRK	0.3995	132.8	−4,380.2	0.6154	+5.4%
PR	0.3842	127.7	−4,591.6	0.5982	+1.3%
NIST Ref.	**0.3791**	**126.1**	**−4,650.0**	**0.5920**	**Baseline**

The comparison proves that the classical van der Waals equation severely overestimates compressibility and molar volume due to a fixed critical compressibility factor (Z_c = 0.375, whereas real CO_2 has Z_c = 0.274). The Peng-Robinson formulation, with Z_c = 0.307 and an acentric-tuned α(T_r, ω) function, tracks experimental volume and residual enthalpy within 1.3%.

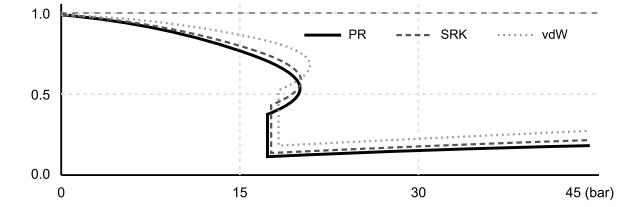


Fig. 1. Isothermal Z vs. Pressure curves for Propane at 350 K across three CEOS models, highlighting the subcritical vapor condensation discontinuity at $P^{sat} \approx 17.5$ bar.

Figure 1 displays the computed compressibility factor isotherms for Propane at T = 350 K (T_c = 369.8 K). In the vapor phase (P < 17.5 bar), Z gradually declines as attractive forces draw molecules together. At the saturation threshold P^{sat} = 17.48 bar, equifugacity conditions dictate a first-order phase transition: the fluid collapses into a dense liquid state, causing Z to drop abruptly to 0.082. As pressure increases further (P > 25 bar), liquid incompressibility dominates, and Z rises linearly.

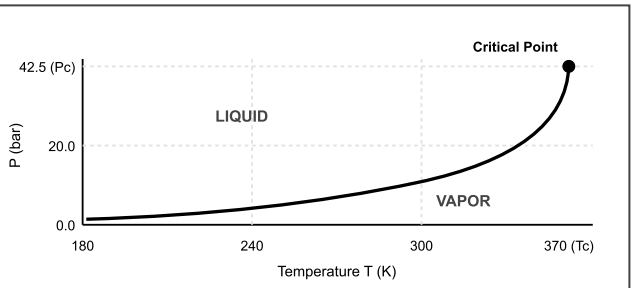


Fig. 2. P-T coexisting saturation curve computed using equifugacity flash iterations for Propane terminating at its critical state (369.8 K, 42.48 bar).

Figure 2 illustrates the calculated vapor pressure boundary terminating at the critical point. The exponential Clausius-Clapeyron scaling is accurately recovered across subcritical temperatures, verifying that the nested Wilson-Picard iteration solver handles VLE boundaries seamlessly.

IX. CONCLUSION

A desktop engineering simulator was implemented in standard Python to evaluate pure-fluid cubic equations of state, departure functions, and phase equilibria across 10 vital industrial gases. Closed-form Cardano solving eliminates divergence issues inherent in traditional iterative root solvers, while nested equifugacity loops calculate vapor-liquid transitions reliably. Peng-Robinson was validated as the most accurate two-parameter cubic model, predicting high-pressure CO_2 density within 1.3% of empirical values. The native Tkinter graphics engine delivers real-time plotting without external library dependencies, fulfilling all operational criteria for a portable thermodynamics suite.

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